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Structure of ^{8}B and astrophysical S_{17} factor

S K Dhiman[1] **and R Shyam**[2]

[1] School of Physical Sciences, Jawaharlal Nehru University, New Delhi 110 067, India
[2] Saha Institute of Nuclear Physics, Kolkata, 700 064, India

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Abstract
Nuclear structure data are of crucial importance in order to address important astrophysical problems such as the origin of chemical elements, the inner working of our sun and the evolution of stars. We demonstrate this by investigating the ground-state structure of ^{8}B and ^{7}Be nuclei within the Skyrme Hartree–Fock framework and by calculating the overlap integral of ^{8}B and ^{7}Be wavefunctions. The latter is used to calculate the astrophysical S factor (S_{17}) for the solar fusion reaction ^{7}Be(p, γ)^{8}B.

1. Introduction

Nuclear astrophysics research addresses some of the most fundamental questions in nature, e.g., the origin of the elements that make up our bodies and our world, and formation and evolution of the sun, the stars and the galaxies. There is an intimate connection between nuclear physics inputs and studies of these fascinating astrophysical phenomena [1]. A diverse set of nuclear data is required to model the composition, changes and energy release in astrophysical environments ranging from the big bang to inner working of our own sun to exploding stars. Theoretical studies as well as measurements of microscopic nuclear physics processes provide the foundation for models to understand various astrophysics phenomena. These models are now challenged by incredibly detailed observations from ground-based and space-based (e.g., CHANDRA X-ray Observatory, Hubble Space Telescope) observational devices that give us an unprecedented view of the cosmos.

We shall demonstrate the role of the nuclear structure physics in nuclear astrophysics by considering the case of the solar fusion reaction ^{7}Be(p, γ)^{8}B, which is of great importance to the solar neutrino issue and to other related astrophysical studies. ^{8}B is the source of the high-energy neutrinos from the sun that are detected in the SNO, Kamiokande and Homestake experiments [2–4]. It is, therefore, a crucial requirement to determine as accurately as possible the cross section of this reaction at relative energies corresponding to solar temperatures (about 20 keV). In this energy region, the cross section $\sigma_{p\gamma}(E_{\mathrm{c.m.}})$ (which is usually expressed in terms of the astrophysical $S_{17}(E_{\mathrm{c.m.}})$ factor) of the ^{7}Be(p, γ)^{8}B reaction is directly proportional to the high-energy solar neutrino flux. A better knowledge of S_{17} is,

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Table 1. rms mass (r_m), proton (r_p) and neutron (r_n) radii for various nuclei. Theoretical results obtained with modified SLy4 Skyrme force. In column 'Experiment' are the values of corresponding radii extracted by fitting the reaction or interaction cross sections by different theoretical methods as discussed in the text. We have defined $r_i = \langle r_i^2 \rangle^{1/2}$

	rms radii (fm)					
	Experiment			Theory		
Nucleus	r_m	r_p	r_n	r_m	r_p	r_n
^{7}Be	2.33 ± 0.02	–	–	2.49	2.63	2.29
^{7}B	–	–	–	2.86	3.18	1.84
^{8}Li	2.37 ± 0.02	2.26 ± 0.02	2.44 ± 0.02	2.54	2.29	2.67
^{8}B	2.55 ± 0.08	2.76 ± 0.08	2.16 ± 0.08	2.57	2.73	2.27
	2.43 ± 0.03	2.49 ± 0.03	2.33 ± 0.03			
^{9}C	2.42 ± 0.03			2.59	2.77	2.20

therefore, important to improve the precision of the theoretical prediction of ^{8}B neutrino flux from the present and future solar neutrino experiments.

$S_{17}(0)$ is determined either by direct measurements [5, 6] or by indirect methods such as Coulomb dissociation [7] and transfer reactions [8, 9]. Efforts have also been made to calculate the cross section of this reaction within the framework of the shell model and the cluster model [10, 11]. The key point of these calculations is the determination of the wavefunctions of ^{8}B states within the given structure theory.

We have investigated the structure of ^{8}B in the framework of the Skyrme Hartree–Fock (SkHF) model which has been used successfully to describe the ground-state properties of both stable [12, 13] as well as exotic nuclei [14]. The SkHF method with density-dependent pairing correlation and SLy4 interaction parameters has been successful in reproducing the binding energies and rms radii [14] in the light neutron halo nuclei ^{6}He, ^{8}He, ^{11}Li and ^{14}Be. We solve spherically symmetric Hartree–Fock (HF) equations with SLy4 [15] Skyrme interaction which has been constructed by fitting the experimental data on radii and binding energies of symmetric and neutron-rich nuclei. Pairing correlations among nucleons have been treated within the BCS pairing method. We have, however, renormalized the parameter of the spin–orbit term of the SLy4 interaction so as to reproduce the experimental binding energy of the last proton in the ^{8}B nucleus [16]. A check on our interaction parameters was made by calculating binding energies and rms radii of ^{7}Be, ^{7}B, ^{8}Li and ^{9}C nuclei with the same set where a good agreement is obtained with corresponding experimental data. The overlap integral of the HF wavefunctions for ^{7}Be and ^{8}B ground states has been used to calculate the astrophysical S_{17} factor.

2. Results and discussions

2.1. Structure calculations

The values of various parameters of the SLy4 Skyrme effective interaction as used in our calculations are given in [16]. The rms radii for matter (r_m), neutron (r_n) and proton (r_p) distributions are presented in table 1 for five light nuclei. Also shown in this table are the matter, neutron and proton radii (in the column 'Experiment') extracted by methods in which measured reaction (or interaction) cross sections are fitted by theoretical models having them as input parameters. The quantities listed in the 'Theory' column are the results of our calculations. Here r_m is obtained by summing the average of proton and neutron radii in every orbit weighted with occupation probabilities. We see that for all the isotopes the calculated r_m is in good agreement with the corresponding values listed in the 'Experiment' column.

2.2. Valence proton radius in ^{8}B and astrophysical S_{17} factor

We define a overlap function of the bound state wavefunctions of two nuclei B and A, where $B = A + p$ (p represents a proton) as

$$I_A^B(\mathbf{r}) = \int \mathrm{d}\xi\, \Psi^*_{AI_AM_A}(\xi)\Psi_{BI_BM_B}(\mathbf{r}, \sigma_p, \xi), \tag{1}$$

where I_A and I_B are the total spins of nuclei A and B, respectively, and $\mathbf{r}$ is the position of the proton with respect to the c.m. of nucleus A. σ_p is the spin variable of the proton, and ξ stands for the remaining set of internal variables which also include isospins. In this expression, nuclear wavefunctions Ψ_A and Ψ_B are supposed to be properly translational invariant. The Hartree–Fock wavefunctions calculated in the previous section may not be so despite the fact that a c.m. correction factor has been incorporated in the energy functional. Corrections for the spurious c.m. motion should, therefore, be applied to the HF wavefunctions before using them in equation (1). However, effects of such corrections on the one-particle overlap function calculated within the shell model have been found [17, 18] to be of the order of only about 2–5%. It is quite likely that the situation will be no different for the HF case.

We now present results for the astrophysical S_{17} factor calculated using HF overlap functions. In the region outside the core where the range of the nuclear interaction becomes negligible, the radial overlap wavefunction of the bound state can be written as

$$R^B_{A\ell j}(r) \simeq \bar{c}_{lj} W_{\eta, l+1/2}(2kr)/r, \tag{2}$$

where W is the Whittaker function, k is the wave number corresponding to the single proton separation energy and η is the Sommerfield parameter for the bound state. In equation (2), $\bar{c}_{lj}$ is the asymptotic normalization constant, required to normalize the radial overlap wavefunction to the Whittaker function in the asymptotic region. The S_{17} factor is related to the proton capture cross section as

$$S_{17}(E) = \sigma(E) E\, \mathrm{e}^{(2\pi\eta(E))}. \tag{3}$$

At the zero energy the S_{17} factor depends only on $\bar{c}_{lj}$ and one can write [10, 11]

$$S_{17}(0) = \kappa \sum_j \bar{c}^2_{1j}. \tag{4}$$

In [10], κ (=37.8 eV b fm) has been obtained by using a microscopic cluster model for the scattering states, while a value of 36.5 eV b fm has been reported for this quantity in [11] using a hard sphere scattering state model. However, once the relevant integration distances are sufficiently enhanced in [11] the value of κ there comes out to be 37.2 eV b fm, which is in good agreement with that of the microscopic model.

In our calculations, we have used our HF overlap function directly in the calculation of the amplitudes for the direct capture reaction. The amplitude for the radiative capture reaction $b + c \to a + \gamma$ can further be written as

$$M_{fi} = \langle \phi_a(\xi_b, \xi_c, r) | \hat{O}(r) | \phi_b(\xi_b)\phi_c(\xi_c)\psi^{(+)}_{k_i}(\mathbf{r}) \rangle, \tag{5}$$

where ϕ, ξ and $\mathbf{r}$ are the wavefunction, the internal coordinate of the bound particle and the relative coordinate between b and c, respectively. $\hat{O}(r)$ is the electromagnetic operator.

Table 2. SkHF results for asymptotic normalization coefficients ($\bar{c}_{\ell j}$), and astrophysical S-factors $S_{17}^{A}(0)$ and $S_{17}^{B}(0)$ and $S_{17}^{C}(0)$. S_{17}^{A} corresponds to results obtained by using the HF overlap function directly to a capture code while S_{17}^{B} and S_{17}^{C} to those obtained by using equation (4) with $\kappa = 36.5$ and 37.8 eV b fm, respectively.

Observable	Our result
$\bar{c}_{13/2}$ fm$^{-1/2}$	0.64
$\bar{c}_{11/2}$ fm$^{-1/2}$	0.34
$S_{17}^{A}(0)$ (eV b)	22.0
$S_{17}^{B}(0)$ (eV b)	19.5
$S_{17}^{C}(0)$ (eV b)	20.2

$\psi_{k_i}^{(+)}(\mathbf{r})$ is the distorted wave in the initial $b + c$ channel. The overlap of initial and final bound state wavefunctions is defined by

$$I_{bc}^{a}(\mathbf{r}) = \langle \phi_a(\xi_b, \xi_c, r) | \phi_b(\xi_b) \phi_c(\xi_c) \rangle,$$
$$= \sum_{\ell j \mu} \langle I_b M_b j m | I_a M_a \rangle \langle \ell m - \mu s \mu | j m \rangle i^{\ell} R_{b \ell j}^{a}(r) Y_{\ell m - \mu}(\hat{r}) \chi_{s\mu}. \qquad (6)$$

It may be noted that at low relative energies the distortion effects are governed solely by Coulomb interactions. Substituting equation (6) into equation (5), expanding $\psi_{k_i}^{(+)}(\mathbf{r})$ in terms of the partial waves and carrying out the angular momentum algebra, the amplitude M_{fi} can be written in terms of the wavefunction $R_{b\ell j}^{a}(r)$ and the radial coulomb wavefunctions for the initial channel. Therefore, the overlap functions $R_{b\ell j}^{a}(r)$ calculated in the HF theory can be used to calculate the capture cross sections at low relative energies without any uncertainty of other input quantities. It may further be noted that using equation (2), we can also calculate the asymptotic normalization constants $\bar{c}_{\ell j}$.

In table 2, we show the results for the asymptotic normalization constants for the $p_{1/2}$ and $p_{3/2}$ proton orbits and the astrophysical S factor obtained by this method. Our $\bar{c}_{3/2}$ agrees almost perfectly with the value of this quantity determined experimentally in [19]. However, our $\bar{c}_{1/2}$ is larger than that reported by these authors by a factor of about 1.5. They have determined these quantities (by invoking the mirror symmetry) from the ANCs of the ^{8}Li $\rightarrow$ ^{7}Li + n system extracted from the measurement of the neutron transfer reaction ^{13}C(^{7}Li, ^{8}Li)^{12}C. This difference in the calculated and 'measured' values of $\bar{c}_{1/2}$ is significant as a larger $\bar{c}_{1/2}$ leads to a bigger S_{17} as compared to that reported in [19]. There may be a need to relook in the theoretical analysis of the data reported in [19]. In the DWBA calculations of the ^{13}C(^{7}Li, ^{8}Li)^{12}C reaction, same set of potentials have been used for ^{7}Li + C and ^{8}Li + C channels. However, ^{8}Li is an unstable system; thus the optical potential in the final channel could be different from that of the incoming (^{7}Li) channel. It is also not very clear from [19] if the spin–orbit terms have been used in the optical potentials and bound state potentials. This could affect the $p_{1/2}$ and $p_{3/2}$ components differently. More work is clearly needed to clarify this issue.

The recent ^{7}Be(p, γ) measurements yield values of $S_{17}(0)$ which are clustered around 18.5 eV b [5] and 22.0 eV b [6]. Our $S_{17}(0)$ (22.0 eV b) is closer to two latest direct capture measurement results which claim a good accuracy. However, this is larger than the most recent result reported by the Texas A&M group from the ANC method [19]. This is also slightly larger than the values obtained from the Coulomb dissociation (CD) method [20], even though it has been argued [21] that there is no significant difference between the CD and direct capture values of $S_{17}(0)$.

3. Summary and conclusions

In summary, in this paper we studied the structure of ^{8}B and ^{7}Be nuclei within the Skyrme Hartree–Fock (SkHF) framework. We calculated binding energies, various densities distribution and rms radii for these nuclei. Using the same set of force parameters, we obtain good agreement with experimental values of binding energies and rms matter radii for all these nuclei. We have calculated the overlap function $\langle ^7\mathrm{Be}|^8\mathrm{B}\rangle$ from the SkHF wavefunctions which has been employed to extract the asymptotic normalization coefficients for the $^8\mathrm{B} \rightarrow {}^7\mathrm{Be} + p$ system. We obtain an astrophysical S factor of 22.0 eV b which lies within the adopted limits ($19.1^{+4.0}_{-1.0}$ eV b) of the near zero energy astrophysical S factor. Our work clearly shows that a proper nuclear structure input is vital in order to understand some very important issues in nuclear astrophysics.

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